
Lab 1: Traffic control digital system

Goals

Learn how to design complex digital systems.

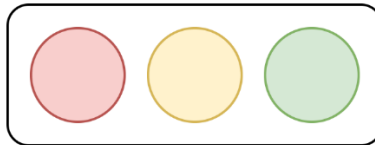
Introduction

All the digital systems that we use daily, consists of very complex digital circuitry consisting typically several computing units and one or more control units.

Lab description

Considering that we are required to design a digital system for traffic control with the following specifications.

- GREEN light must last 10 clock cycles.
- YELLOW light must be on 2 clock cycles then off 2 clock cycles and then on 2 clock cycles.
- RED light must last 15 clock cycles.
- The full cycle must start with the GREEN light at the power up.
- The cycle must start cyclically once an input signal toggles high, the cycle must stop until another input signal toggles high.



Submission

Implementation description

Generate a document providing a detailed description of the implementation. This should include an explanation of design decisions, challenges encountered, and how they were overcome. Include micro architecture diagrams, schematics, tables, or equations used in design development. Also, provide some snaps of the simulation of the digital system, the snap should clearly show the full sequence of the traffic control lamps, this report should also include the RTL code.

Answer

Use your imagination 😊